MIL-C-22653C(AS)
10 December 1985
SUPERSEDING
MIL-C-22653B(Wep)
27 October 1964

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 19 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 19 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering

Beneficial comments (recommendations, additions , deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1377

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2.1.2 **Drawings and publications.** The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

1825934	Igniter Assembly
2015079	Loaded Spacer
DL419700	Cartridge, Impulse, Mark 19 Mod 0
DL537883	Container, Inner, for Cartridge Impulse, Mark 19
DL537884	Container, Outer, for Cartridges, Power Source

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5	Ammunition and Explosives Ashore
WR-43	Preparation of Quality Assurance Provisions

Code of Federal Regulations

49 CFR 100-199	Transportation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 **Order of Precedence.** In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 **Conformance to documents.** The Mark 19 Mod 0 impulse cartridge covered by this specification shall be manufactured in accordance with DL419700 and all documents listed thereon.

3.3 **Materials.** All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

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3.4 Primary components. For the purposes of this specification the propellant is considered a primary component.

3.5 Insulation resistance. Prior to the assembly of the bridgewires in the inner cup and electrode assembly the insulation resistance shall be not less than 50 megohms at a relative humidity of less than 80 percent (see 4.6.1).

3.6 Bridgewire. The bridgewire shall be soldered to the inner cup and electrode. The solder and soldering process shall be as specified on the applicable drawing.

3.7 Bridgewire assembly, continuity. Before filling the cavity around the bridgewire with the igniter compound, the bridgewire assembly shall be thoroughly cleaned to remove solder flux and contamination. After cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of both legs of the bridgewire. The current shall be regulated to produce a visible red glow of the bridges. An electromagnetic force (EMF) of approximately 1.5 volts ac or dc may be used with a suitable series resistor to adapt the source characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to flow in both legs of the bridge shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge.

3.8 Igniter composition and loading. The igniter composition shall be compounded and loaded in accordance with Drawing 1825934.

3.8.1 Loaded spacer. The loaded spacer shall be loaded with primer composition FA-878 in accordance with Drawing 2015079.

3.9 Propellant charge.

3.9.1 Quantity. All propellant charges shall be as specified on Drawing 2164465. Charges shall be determined by weight rather than by volume measurement.

3.9.2 Smokeless powder, handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.9.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.6.2).

3.10 Cartridge requirements.

3.10.1 Bridgewire resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.6.4), shall be not less than 0.08 ohm nor more than 0.14 ohm.

3.10.2 Leakage. Cartridge leakage rate shall be not greater than 10^{-5} cubic centimeters of air per second (see4.6.5).

3.10.3 X-ray. Cartridges shall be free of imperfections when X-rayed (see 4.6.6).

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3.11 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.11.1 through 3.11.3 when subjected to the test prescribed in 4.6.7.

3.11.1 Chamber pressure. The upper (U) and lower (L) limits of maximum pressure in pounds per square inch shall be 2400 and 1550, respectively.

3.11.2 Misfire. There shall be no misfires (see 4.6.7.2).

3.11.3 Hangfire. There shall be no hangfires (see 4.6.7.3).

3.12 Change in propellant lot. Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.8. No cartridge shall produce results outside the limits specified in 3.11.1 and no cartridge shall fail to meet the requirements of 3.11.2 or 3.11.3.

3.13 Visual inspection. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, missing or damaged O-ring, perforated closure disc, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.2 through 3.10.3 and 3.13 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the intended sources of supply.

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4.4.1 Sample size. The first article sample shall consist of 94 cartridges.

4.4.2 Inspection routine. Eighty-four cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests as prescribed in table III of MIL-D-21625 (see 6.2): 40-foot drop test (6); 15 "g" shock (12); THC (18); vibration (12); −65°F temperature (12); +160°F temperature (12); +70°F temperature (12). The number in parentheses denotes the number of cartridges allocated for each test. The cartridges shall be fired in the test set specified in 4.6.7. No cartridge firing shall produce results outside the limits specified in 3.11.1 through 3.11.3.

4.4.3 Failure. Failure of any cartridge shall be cause for rejection of the first article sample.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table I.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement para.	Test para.	Classification (WR-43)
Insulation	Resistance	3.5	4.6.1	(M101)
Propellant charge	Quantity	3.9.1	-	(M102)
Smokeless powder	Condition	3.9.3	4.6.2	(M103)
Cartridge	Visual examination	3.10.4	4.6.3	(M104)
Cartridge	Bridgewire resistance	3.10.1	4.6.4	(M105)
Cartridge	Leakage	3.10.2	4.6.5	(M106)
Cartridge	X-ray	3.10.3	4.6.6	(M107)
Cartridge	Ballistics	3.11	4.6.7	(M108)
Cartridge	Preservation-packaging, packing and marking	Section 5	4.6.3	(M109)

4.5.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 1301 and not greater than 8000 maximum. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.5.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
66 - 110	15	10
111 - 180	15	10
181 - 300	20	10
301 - 500	25	10
501 - 800	30	10
801 - 1300	35	10
1301 - 3200	40	10
3201 - 8000	50	10
8001 - 22000	75	10

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4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Tests.

4.6.1 Insulation resistance. Prior to assembly of the bridgewire in the inner cup and electrode assembly the insulation resistance shall be checked with a 500-volt megohmmeter. Units with resistance of less than 50 megohms shall be cause for rejection.

4.6.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.6.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.13 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5.

4.6.4 Bridgewire resistance. Conformance with the bridge circuit resistance of 3.10.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.10.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.6.5 Leakage. Each cartridge in the first article or production sample shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate greater than 10^{-5} cc per second of air at a pressure differential of 1 ± 0.1 atmosphere shall be considered defective. Cartridges of the first article sample which exhibit a leakage defect shall be cause for rejection. If the number of cartridges of the production lot sample exhibiting leak defects exceeds an AQL of 2.5 percent defective from table II-A of MIL-STD-105, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the Government. Cartridges which exhibit a leak rate in excess of that specified above shall be cause for rejection (see 6.5).

4.6.6 Radiographic examination. All cartridges shall be radiographically examined in accordance with MIL-STD-453. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.6.7 Ballistic tests. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for 24 hours at 70° ± 5°F and fired in a closed bomb test fixture having a volume of 2.7 cubic inches (including the cartridge volume). The

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chamber shall be thoroughly cleaned before each firing and the cartridge temperature shall be 70°F. A record of pressure versus time (i.e. time from application of the firing current) shall be taken for each firing. Firing current shall be 10 amperes. Acceptance criteria (regardless of lot size) shall be: accept zero defects and reject if one or more defects.

4.6.7.1 Criteria of acceptability. The upper (U) and lower (L) limits of maximum pressure shall meet the requirements of 3.11.1. A failure of any cartridge to meet this requirement shall be cause for rejection of the lot represented.

4.6.7.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.6.7.3 Hangfire. One or more hangfires (see 6.4.5) shall be cause for rejection of the lot represented.

4.7 Retest. There shall be no retests.

4.7.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

4.8 Change in propellant lot. Whenever a change in propellant lot for the main charge occurs, the charge weight shall be verified by firing 36 cartridges which have been temperature conditioned as follows: 12 at −65° ± 5°F, 12 at 70° ± 5°F, and 12 at 160° ± 5°F. Results of these firings shall meet the requirements of 3.12.

5. PACKAGING

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with DL537883.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in first article sample testing and production lot acceptance testing shall be packaged in accordance with method III of MIL-P-116 and with 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with DL537884.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When specified in the contract (see 6.2), cartridges intended for expenditure in first article testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with 49 CFR 171-178.

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5.3 Marking. Cartridge marking shall be as specified on Drawing 2164465 and MIL-STD-1168. In addition, packaging and packing markings shall be in accordance with 49 CFR 171-178, MIL-STD-129, and MIL-STD-1168.

6. NOTES

6.1 Intended use. The Mark 19 Mod 0 impulse cartridge is intended primarily as an aircraft stores separation cartridge (replacing the Mark 25 primer).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required, and if so, specify the test activity
c. Date and place of delivery
d. Inspection conditions, if other than as specified (see 4.3)
e. That revision "C" of MIL-D-21625 be used (see 4.4.2)
f. Lot size, if other than as specified (see 4.5.1)
g. Sampling, if other than as specified (see 4.5.2)
h. Special marking, if any (see 5.3)
i. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5 are applicable.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 94 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. For the purpose of this specification, the following definitions apply.

6.4.1 Level A packaging. The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.2 Level C packaging. The degree of preservation-packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.4.3 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

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6.4.4 Level C packing. The degree of packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.4.5 Hangfire. A hangfire is defined as a firing in which the elapsed time between actuation of the fir ing mechanism and case opening or start of pressure rise exceeds 0.05 second.

6.5 Advisory leak test. Indian Head leak tests cartridges in accordance with "Cartridge Leak Test Pro cedures at the Naval Ordnance Station, Indian Head, Md."

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22653C(AS)	CARTRIDGE, IMPULSE, MARK 19 MOD 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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